

# PAX vibration diagnostics — run report

## RUN

235148\_pax-vibration\_pax-vibration

Status: completed | Started: 2026-09-24T22:51:48.099910+00:00

## PURPOSE

Compare fast PD variance with the PAX motor running and stopped. Log all fresh PAX telemetry and static sphere-angle variance during the on window. Duration is per condition; no actuator waveform.

## REQUIRED OPTICAL SETUP

PD on selected IN1/IN2 with DC coupling and unchanged light/gain in both conditions. Keep PAX physically mounted and optics fixed. Motor-off means rotation stopped, not power unplugged. Outputs hold the configured static biases (default zero).

## RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

## OPERATOR SETUP NOTES

V<sub>pi</sub> expectation only; verify channel and measured driver gain before calibration.

## DATA

52 recorded samples; pax\_on: 25, pax\_off: 27

Source: data.csv; full acquisition settings: recipe.json; status: run.json

# PAX motor noise — variance, band [5.0, 1000.0] Hz

## PAX\_ON

sample\_count: 409600  
acquired\_s: 26.843545599999995  
wall\_span\_s: 30.297339292999823  
sampling\_rate\_hz: 15258.789062499998  
dark\_corrected\_mean\_v: 0.012215372025966644  
normalized\_variance: 0.0006465075815047131  
normalized\_band\_variance: 1.8658136785781005e-05  
status: ok

## PAX\_OFF

sample\_count: 442368  
acquired\_s: 28.991029247999993  
wall\_span\_s: 30.385372937016655  
sampling\_rate\_hz: 15258.789062499998  
dark\_corrected\_mean\_v: 0.012237156154932798  
normalized\_variance: 0.0002304015752699564  
normalized\_band\_variance: 1.148121795304651e-05  
status: ok

## COMPARISON

normalized\_variance\_on\_minus\_off: 0.0004161060062347567  
normalized\_variance\_on\_over\_off: 2.806003304218795  
normalized\_band\_variance\_on\_minus\_off: 7.176918832734496e-06  
normalized\_band\_variance\_on\_over\_off: 1.6251008265921927  
interpretation: Signed on-minus-off excess, not proof of mechanical causation. Compare the same PD bandwidth, optical alignment, light level and gain. Sequential drift, shot noise and electronics can also differ. PAX power/angles have a much slower bandwidth.

## PAX ON: STATIC POLARIZATION / POWER

sample\_count: 200  
mean\_power\_w: 0.0002948098815

# PAX motor noise — variance, band [5.0, 1000.0] Hz

normalized\_power\_variance: 0.005725433791958411

u\_variance\_rad2: 0.0269251164131837

v\_variance\_rad2: 0.051680362060569376

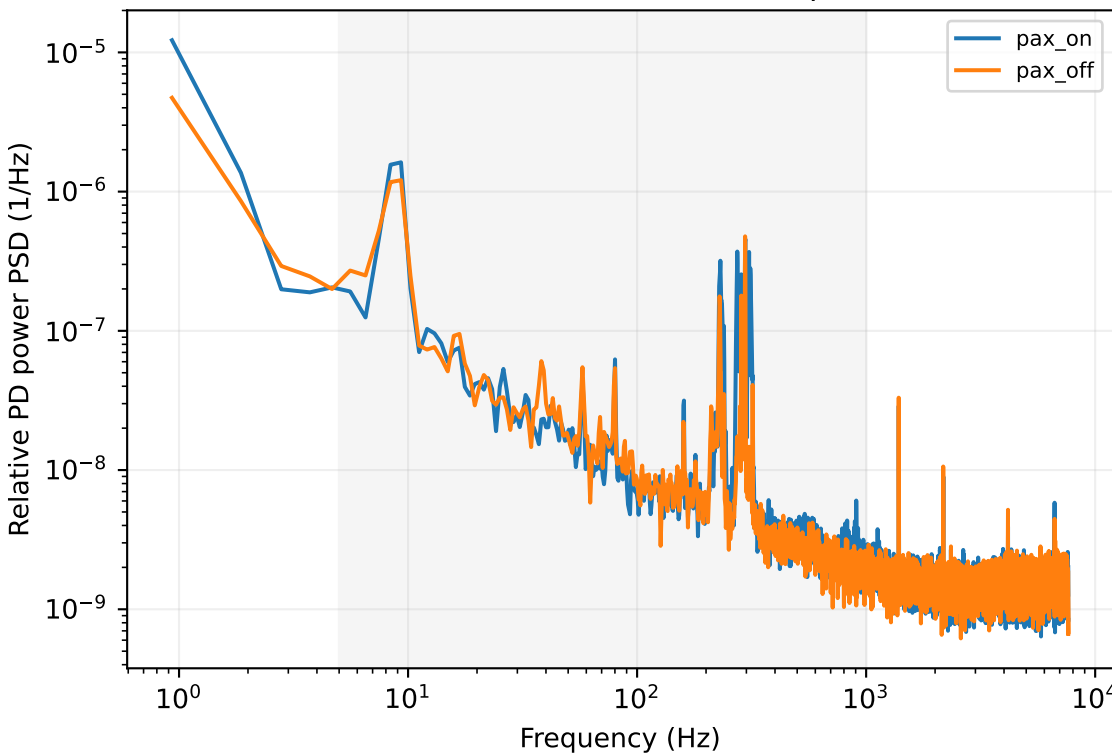
mean\_dop: 0.8728819515

minimum\_sin\_v: 0.14308558254600298

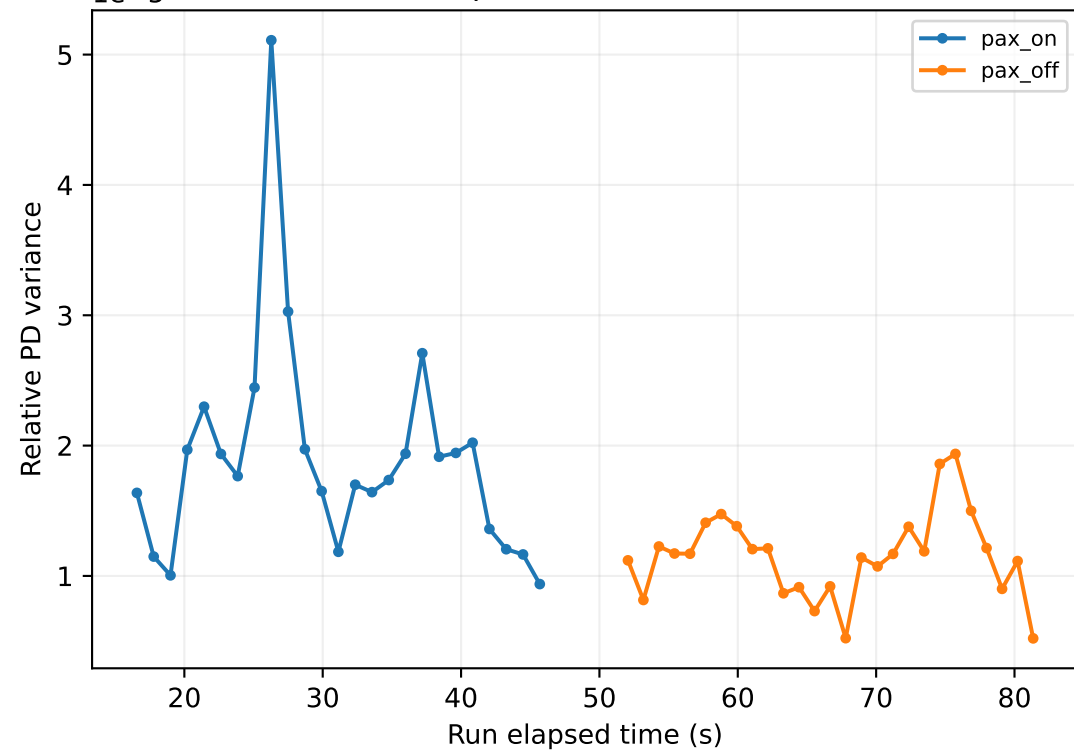
angle\_note: u unwrapped in acquisition order; coordinate noise grows near S1 poles

update\_rate\_hz: 6.603824011022595

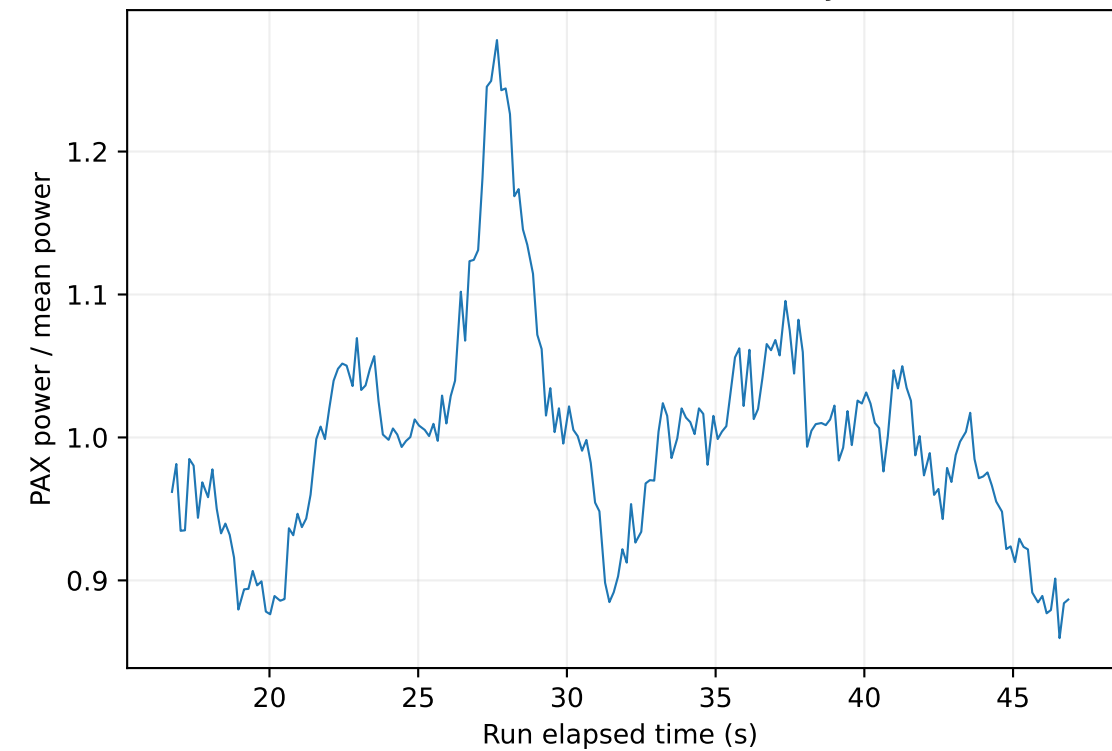
Identical PD bandwidth; shaded comparison band



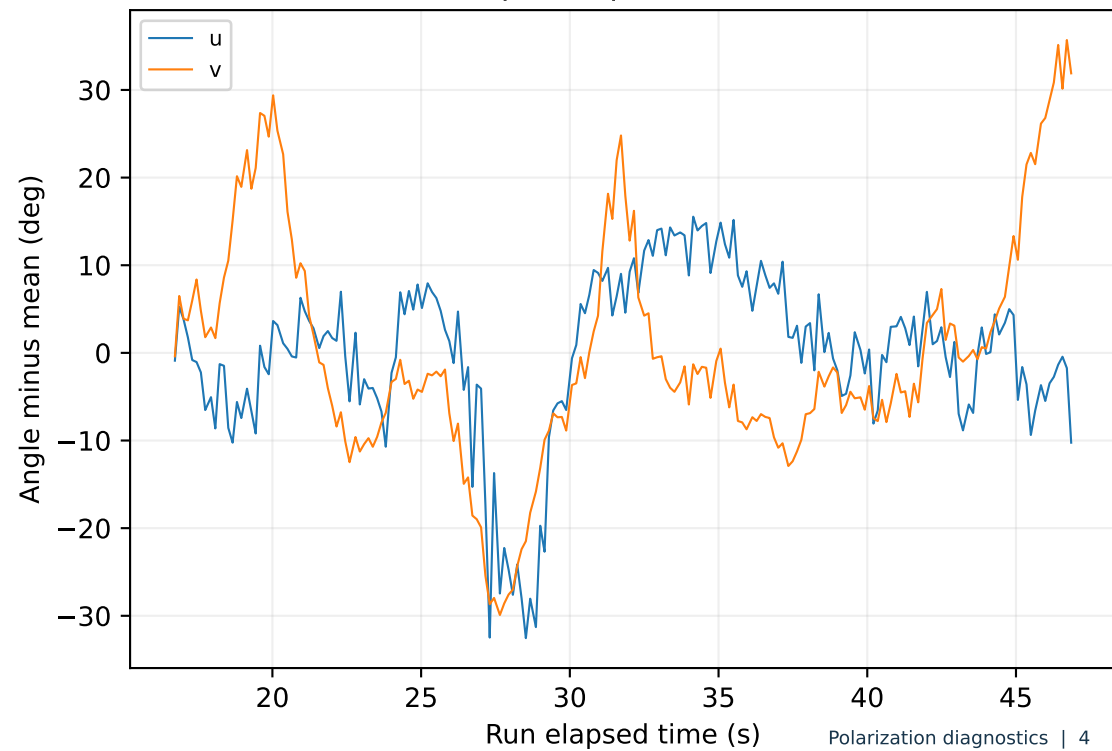
Per-capture band variance



PAX on: slow reference only



Static S1-polar sphere coordinates



# Active parameters

RP commands: volts, bounded to 0–1. V\_lambda: terminal volts for 2π. Gains: terminal V / RP V.

| Parameter                       | Recorded value  |
|---------------------------------|-----------------|
| label                           | "pax-vibration" |
| duration_s                      | 30.0            |
| report                          | "pdf"           |
| pd_input                        | "in2"           |
| pax_rotation_velocity_hz        | 60.0            |
| pax_measurement_mode            | 5               |
| pax_wavelength_nm               | 830.0           |
| pax_vibration_scope_decimation  | 8192            |
| pax_vibration_band_low_hz       | 5.0             |
| pax_vibration_band_high_hz      | 1000.0          |
| pax_vibration_motor_settle_s    | 5.0             |
| pax_vibration_dark_voltage_v    | null            |
| pax_vibration_phi1_bias_voltage | 0.0             |
| pax_vibration_phi2_bias_voltage | 0.0             |